B.Tech CSE - Third Year | Lab 1

Assignment 1: Reading Array Values • *Aim: Read and process array values using indexes and loops.*

Experiment No. 1.1: Read first and last array values

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Read_first_and_last_array_values {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        System.out.println("First: " + marks[0]);
        System.out.println("Last: " + marks[marks.length - 1]);
    }
}
```

3. **Output:**

```
First: 45
Last: 90
```

4. **Viva-Voce:** What is an array index? What is the first valid index in Java? How do you find the last index?

Experiment No. 1.2: Print all array values

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Print_all_array_values {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        for (int i = 0; i < marks.length; i++) {
            System.out.println(marks[i]);
        }
    }
}
```

3. **Output:**

```
45
67
80
52
90
```

4. **Viva-Voce:** What does marks.length return? Why does the loop start at 0? What is a loop iteration?

Experiment No. 1.3: Calculate array sum

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Calculate_array_sum {
    public static void main(String[] args) {
        int[] marks = {45, 67, 80, 52, 90};
        int sum = 0;
        for (int value : marks) {
            sum += value;
        }
        System.out.println("Sum: " + sum);
    }
}
```

3. **Output:** `Sum: 334`
4. **Viva-Voce:** What is an accumulator? What is an enhanced for loop? Why initialize sum to 0?